

# Directly converting Fe-doped metal–organic frameworks into highly active and stable Fe–N–C catalysts for oxygen reduction in acid

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## ABSTRACT

Compared to extensively studied oxygen reduction reaction (ORR) catalysis in alkaline media, development of highly active and stable nonprecious metal catalysts (NPMCs) to replace Pt in acidic electrolytes remains grand challenges. Among currently studied catalysts, the Fe–N–C formulation holds the greatest promise for the ORR in acid. Here, we report a new highly active and stable Fe–N–C catalyst featured with well-dispersed atomic Fe in porous carbon matrix. It was prepared through one single thermal conversion from Fe-doped ZIF-8, a metal–organic framework (MOF) containing  $\text{Zn}^{2+}$  and well-defined  $\text{Fe-N}_4$  coordination. Unlike other Fe–N–C catalyst preparation, no additional tedious post-treatments such as acid leaching and the second heating treatment are required in this work. Notably, an  $\text{O}_2$ -free environment for preparing the Fe-doped ZIF-8 precursor is found to be crucial for yielding uniform Fe distribution into highly porous N-doped carbon matrix. The resulting new Fe–N–C catalyst exhibited exceptionally improved ORR activity with a very high half-wave potential (0.82 V vs. RHE) and sufficient potential cycling stability in acid. Opposite to previous observation, the highly active Fe–N–C catalyst is in the absence of any graphitized nanocarbons, which would lead to a new discussion in the field for understanding the role of carbon during the ORR electrocatalysis.

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## 1. Introduction

Commercialization of polymer electrolyte fuel cells (PEFCs) for transportation and other potential applications is largely hindered by their high cost and insufficient performance durability [1–3]. The substantial cost is due to the large amount of Pt-based catalysts used at the cathode to catalyze the oxygen reduction reaction (ORR), a sluggish electrochemical reaction [4–6]. Developing nonprecious metal catalysts (NPMCs) derived from earth abundant elements for the ORR, therefore, is of critical to address the cost issue in PEFCs [7–14]. Since the first discovery of ORR activity of the Co phthalocyanine complex in 1964 [11], the heat treated M–N–C (M = Fe or Co) ORR catalysts have been extensively studied [10,15]. Such catalysts are typically prepared via pyrolyzing a composite precursor containing iron or cobalt, nitrogen, and carbon at temperatures above 700 °C to achieve high activity and a robust structure [1,14,16–19]. The speculated ORR active sites are

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associated with N coordinated metal structures ( $\text{MN}_x$ ) with average coordination number  $x$  from 2 to 4 evidenced by X-ray absorption techniques or Mössbauer spectroscopy [20–22]. Oppositely, an electrocatalyst with an active site devoid of any direct nitrogen coordination to iron recently exhibits comparable performance to the best performing Fe–N–C catalysts in acidic media [23]. In-situ X-ray absorption spectroscopy combined with advanced electron microscopy indicated that iron is not directly involved in the oxygen reduction pathway and is embedded into nitrogen-doped carbon fibers. Instead of iron active sites, the reaction occurs primarily on the carbon–nitrogen structures in the outer skin. Therefore, the nature of active site is still uncertain in the field. Despite the controversial evidences, among studied nonprecious metal catalyst formulations, the Fe–N–C catalyst represents the best NPMC for the ORR in terms of the most promising activity and stability, particularly in acidic electrolytes [24,25].

Although great progress has been made in last decade, the significant gaps between the Fe–N–C and Pt catalysts must be bridged for viable application in terms of their activity and durability especially in harsh acidic electrolytes [3,6,26–30]. A key step to further enhance the performance of the Fe–N–C catalyst is

to increase the density of the ORR active  $\text{FeN}_x$  sites in a robust 3D architecture. Because the catalyst performance is associated with the density of the active sites and their local structural environments [31]. Previous work has indicated that simply raising the Fe concentration in precursors during the synthesis is not effective, as this will lead to aggregation of iron, thereby generating inactive iron compounds [3]. The reported Fe-N-C catalysts with good performance typically contain less than 10 wt% Fe, while inactive aggregated particles of iron species (e.g., metallic iron, oxides or carbide) remain visible in most cases [1,3,6,10,15,16,32,33]. Thus, it remains a grand challenge to uniformly disperse active atomic iron into high-surface area carbon matrix, instead of generating iron agglomerates. Most of currently studied Fe-N-C catalysts are prepared by pyrolyzing a composite that was mixed with iron salts, nitrogen-carbon precursors, and high-surface area supports, followed by a tedious acidic leaching and the second treatment [24]. These current synthetic methods often lead to a heterogeneous morphology without accurate control of local structures for active sites. Exploring a highly ordered and porous precursor is demanded to achieve uniform distribution of active sites into a porous architecture with significantly increased density of active sites [34]. Recently metal-organic frameworks (MOFs), built by secondary building of units from metals and organic linkers with strong bonding, have received a lot of attentions since they possess permanent porosity, three-dimensional (3D) structure and diversity of metals and organic linkers. The structure of MOFs consists of ligands containing carbon and nitrogen atoms coordinating with metals to form ordered 3D porous crystals. These structural and chemical properties not only meet the requirements of M-N-C composition as templates but also provide ordered three dimensional porous structures. Additionally, the strong interaction among metal atoms and nitrogen atoms would be beneficial for generating more active sites and higher intrinsic catalytic active sites during the carbonization. In 2010, Liu initiated the preparation of Co/Fe-N-C catalysts derived from MOFs for ORR cathodes in PEFCs [35–38]. Their promising results started bringing attention to societies, which MOF can be a good carbon/nitrogen precursor used for such catalyst preparation. In 2011, Dodelet et al. [16] developed a new Fe-N-C catalyst through physical mixing of ZIF-8, a zeolitic imidazolate framework (ZIF) composed of Zn and 2-methylimidazole, with N-chelating Fe compounds using ball milling followed by two annealing treatments using Ar and  $\text{NH}_3$ , respectively. The advantage of using ZIF-8 was highlighted in that work. It was found that pyrolysis of ZIF-8 in inert atmosphere can yield highly porous N-doped carbon as Zn is volatile at high temperature and can be eliminated during pyrolysis. The resulting catalyst generated a high power density in fuel cell tests due to significantly increased active site number and facilitated mass transport in the porous catalyst layer. However, the synthesis procedure is still complicated involved tedious ball milling and two-time heat treatments. In addition, Fe precursors is still mixed with MOF by a physical way, rather than well-controlled chemical doping.

In this work, we report a new synthesis method to prepare a highly active Fe-N-C catalyst featured by uniform Fe distribution at an atomic level. During the synthesis, for the first time, Fe was successfully doped into ZIF-8 (zeolitic imidazolate frameworks-8) by partially replacing Zn ions. As a results,  $\text{Fe(II)}$  was uniformly distributed into the framework with well-defined  $\text{Fe-N}_4$  coordination. The highly active and stable Fe-N-C catalyst was prepared by only one-step thermal treatment of Fe-doped ZIF-8. No additional post-treatments such as acid leaching and the second heating treatments is required. During the thermal treatments,

nitrogen and iron containing ZIF-8 framework directly converts into N-doped carbon with uniform Fe distribution without any agglomerated iron particles. The zinc species is eliminated due to its high volatility, yielding a highly porous carbon morphology. Notably, the control of an  $\text{O}_2$ -free environment for stabilizing Fe (II), rather than forming  $\text{Fe(III)}$ , during the preparation of Fe-doped ZIF-8 precursor, was found to be extremely crucial for generating uniform Fe distribution at an atomic level. This strategy effectively prevents formation of inactive iron aggregates. The exceptionally improved ORR activity is benefited from the uniform Fe distribution into porous carbon, capable of providing maximum density of the ORR active Fe sites.

## 2. Experimental details

### 2.1. Catalyst synthesis

Fe-doped ZIF-8 nanocrystals were prepared by using a surfactant mediated growth method. Traditionally, the preparation of ZIF-8 crystals needs much excessive ligands in aqueous solution. Surfactant methods enable the control of particle size of the ZIF-8 and are also crucial to prevent the formation of alkaline salt of Zn (e.g.  $\text{Zn(OH)NO}_3$ ) in aqueous solution. The micelles formed by the two surfactants (Span 80 and Tween 80) will protect the  $\text{Zn}^{2+}$  from attacking by  $\text{OH}^-$  and allows the  $\text{Zn}^{2+}$  preferentially binding to 2-methylimidazole. Span 80 and Tween 80 are often used simultaneously [39,40] to obtain uniform ZIF-8 nanocrystals. Typically, solution #1 contains 1.5 g surfactants (Span 80 and Tween 80, 9:11 in mass ratio), 0.1 mol 2-methylimidazole (MeIM), and 200 mL deionized water. Solution #2 contains 0.95 mmol  $\text{Zn(NO}_3)_2$ , 5 mmol  $\text{FeSO}_4$ , and 100 mL deionized water. Both solutions are stored in round flasks and are bubbled with high purity argon for 1 h. Then solution #2 is mixed with solution #1 through a tube connecting the two flasks using compressed Ar to initial the reaction. The mixed solution is stirred at 60 °C under the protection of Ar for 12 h until a greyish turbid suspension appears. The resulting product is separated by centrifugation, washed by deaerated ethanol and dried at 80 °C in vacuum oven for 12 h. The resulting sample was labeled as Fe-Z8-Ar. For a comparison, the Fe-doped ZIF-8 was also prepared under the identical procedures without Ar protection (Fe-Z8-air). As a result, yellowish powder is obtained. The ORR catalysts are prepared by temperature programmed pyrolysis of the Fe-doped ZIF-8 nanocrystals in an Ar atmosphere. Typically, 0.5 g Fe-doped ZIF-8 nanocrystals was heated to 1000 °C at 5 °C  $\text{min}^{-1}$  under an Ar stream in a tube furnace, then kept at the peak temperature for 60 min and then allowed to cool down to room temperature automatically.

### 2.2. Materials characterization

Structure and morphology of studied samples are characterized by X-ray diffraction (XRD, Rigaku D/max 200 diffractometer,  $\text{Cu K}\alpha$ ), and high-resolution transmission electron microscopy (HRTEM, JEM 2100, 200 kV). Nitrogen adsorption-desorption isotherms are measured on an Autosorb IQ gas sorption analyzer (Quantachrome) at 77 K. Before testing, samples are degassed at 200 °C for 2 h. The surface area and pore size distribution are determined by the Brunauer-Emmett-Teller (BET) and Quenched Solid Density Functional Theory (QSDFT) methods assuming a slit/cylinder pores model, respectively. The X-ray photoelectron spectroscopy (XPS) analysis is performed on an AXIS-Ultra spectrometer (Kratos Analytical) using monochromatic Al  $\text{K}\alpha$  radiation

(225 W, 15 mA, 15 kV). Content of C, H and N is determined by the combustion method on an elemental analyzer (Vario EL). The metal content is obtained by Inductively Coupled Plasma-Atomic Emission Spectrometer (ICP-OES), which is performed on a CCD-ICP-OES spectrometer PROFILE SPEC. The samples are first calcinated in air at 900 °C for 12 h. Then the obtained residue are dissolved in aqua regia to prepare the solution for ICP-OES analysis.

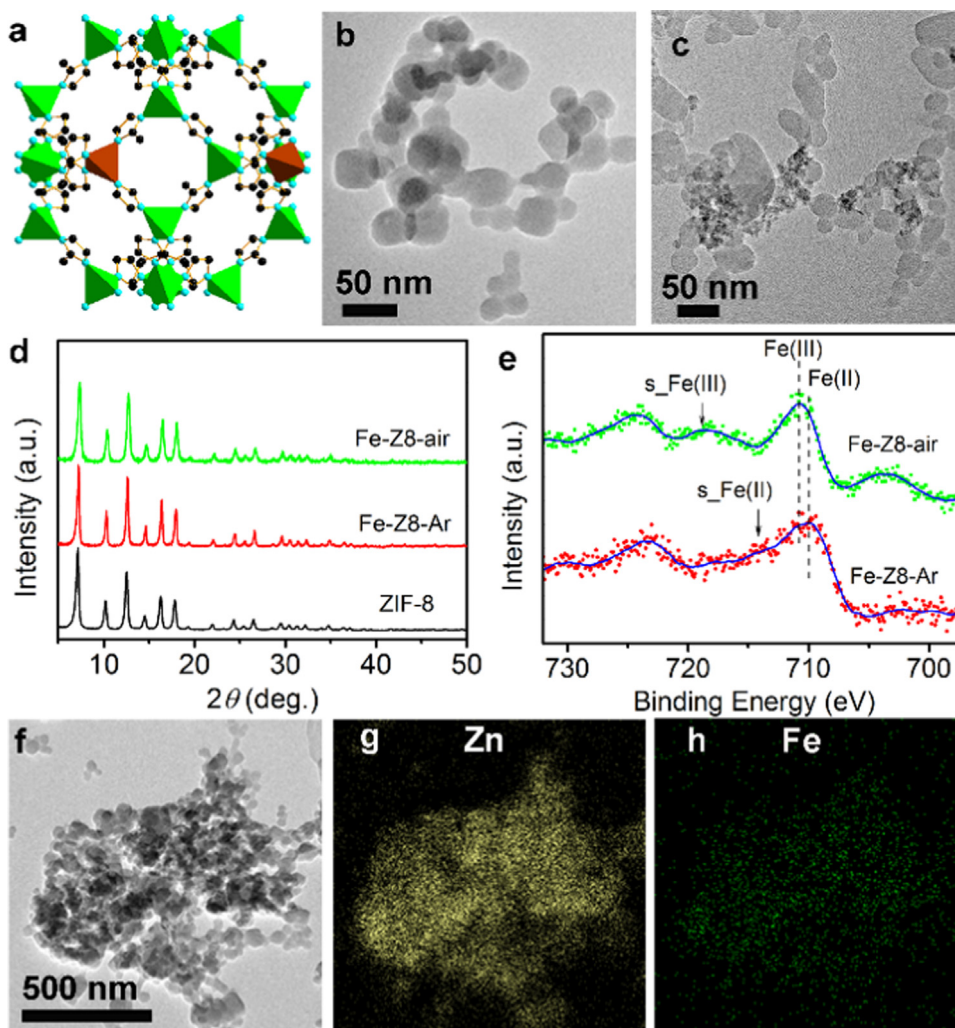
### 2.3. Electrochemical measurements

ORR performance is evaluated in 0.1 M HClO<sub>4</sub> solution on a glassy carbon rotating ring disk electrode (RRDE, 5.61 mm of disk outer diameter, Pine Research Instrumentation, USA). Catalysts are loaded by drop casting isopropanol suspension consisting of catalyst powder and Nafion ionomer. The catalyst loading is controlled at 0.56 mg/cm<sup>2</sup>. A Pt/XC-72 (20 wt%) catalyst with a loading of 20 μgPt cm<sup>-2</sup> was used as a reference. The reference electrode is a saturated calomel electrode (SCE) and calibrated using reversible hydrogen electrode (RHE). All potentials reported in this work have been converted to the RHE scale. The counter electrode

is a graphite rod. The methanol tolerance characterization of the catalysts was carried out in the potential range from 1.2 to 0.0 V at a scan rate of 50 mV s<sup>-1</sup> in O<sub>2</sub>-saturated 0.1 M HClO<sub>4</sub> containing 1 M methanol. The stability tests of the C-Fe-Z8-Ar catalyst was performed in O<sub>2</sub>-saturated 0.1 M HClO<sub>4</sub> in the potential range from 0.6 to 1.0 V at room temperature.

In RRDE experiments, the ring potential was set to 1.2 V. Before experiments, the Pt ring was activated by potential cycling in 0.1 M HClO<sub>4</sub> from 0 to 1.4 V at a scan rate of 50 mV s<sup>-1</sup> for 10 min. ORR steady-state polarization plots were recorded in O<sub>2</sub>-saturated 0.1 M HClO<sub>4</sub> using potential step of 0.03 V and wait-period of 30 s between two subsequent potentials, which allow the non-faraday current pass away. This method can prevent the capacitive current from affecting ORR Faraday current. Thus, it does not need to subtract capacitive background current. The peroxide yield H<sub>2</sub>O<sub>2</sub>% is calculated by the ring current (*I<sub>r</sub>*) and the disk current (*I<sub>d</sub>*) by using the equation [41]:

$$\text{H}_2\text{O}_2\% = \frac{200I_r}{I_r + NI_d} \quad (1)$$



**Fig. 1.** (a) Schematic illustration of the structure of Fe doped ZIF-8. ZnN<sub>4</sub> and FeN<sub>4</sub> tetrahedrons are shown in green and yellow, respectively. (b,c) TEM images of Fe-Z8-Ar (b) and Fe-Z8-air (c). (d,e) XRD patterns (d) and XPS spectra (e) of Fe-Z8-Ar and Fe-Z8-air. The XRD pattern of ZIF-8 is also shown for comparison in (d). The satellite peak in XPS spectra (e) are denoted by the arrows. (f–h) Elemental mapping of Fe-Z8-Ar in the scanning TEM mode of the corresponding region in (f). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



where  $N=0.37$  is the collection efficiency, calibrated by the  $K_3Fe(CN)_6$  redox reaction. The kinetic current density ( $I_k$ ) used for determining mass activity is calculated by the measured current density ( $I_F$ ) and limiting current density ( $I_L$ ) according to the equation [41,42]:

$$I_k = -\frac{I_F I_L}{I_F - I_L} \quad (2)$$

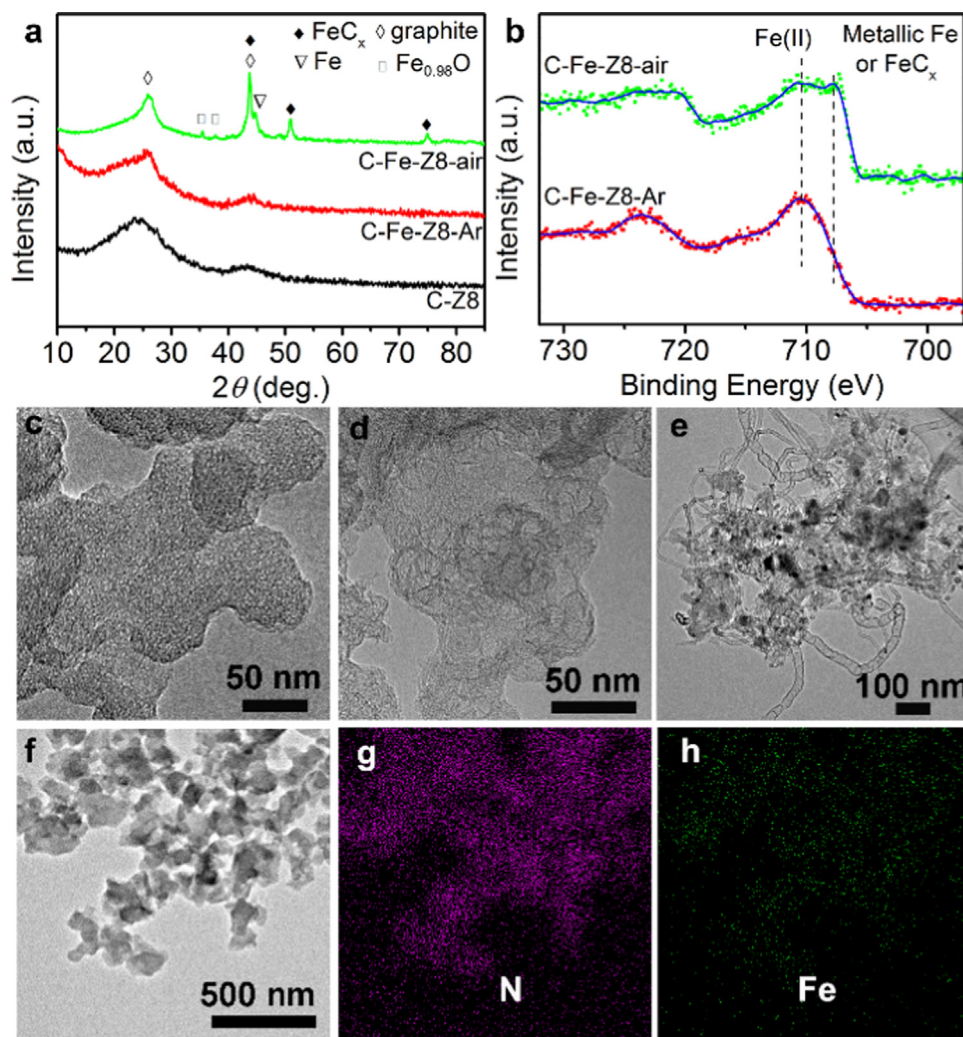
### 3. Results and discussion

#### 3.1. Synthesis of Fe-doped ZIF-8

A nanocrystal structure of Fe-doped ZIF-8 was shown in Fig. 1a. A simple pyrolysis for the Fe-doped ZIF-8 was applied to yield high Fe dispersion on N-doped carbon after Zn elimination. Moreover, the well-defined  $FeN_4$  coordinating structure in the Fe-doped ZIF-8 will facilitate the directly formation of the ORR active  $FeN_x$  sites. Su et al. once attempted Fe doping by introducing Fe(II) during the solvothermal growth of ZIF-8 [43]. However, that approach failed to achieve high Fe dispersion in the resulting catalyst with insufficient ORR activity even in alkaline electrolyte. Alternatively,

ZIF-8 (or similar Zn based ZIFs) once was physically mixed with N-chelating Fe compounds using ball milling, leading to more efficient Fe-N-C catalysts after pyrolysis as demonstrated by Liu et al. [35,36] and Dodelet et al. [16]. Therefore, profound insights into the synthetic process particularly for the interactions between Fe and ZIF-8 framework are certainly needed. In our synthesis, we discovered that an  $O_2$ -free environment is desirable to realize optimal Fe doping directly into ZIF-8, which are crucial for achieving uniform Fe dispersion at atomic level with high ORR activity.

In the newly developed method, Fe-doped ZIF-8 was prepared in an aqueous emulsion containing metal salts ( $FeSO_4 \cdot 7H_2O$  and  $Zn(NO_3)_2 \cdot 6H_2O$  in a molar ratio of 5:95), 2-methylimidazole and surfactants (Span 80 and Tween 80) at 60 °C. After stirring for 24 h, the solid product is collected by centrifugation followed by washing with ethanol. The preparation is carried out in both ambient air and an Ar atmosphere. The obtained products are denoted as Fe-Z8-air and Fe-Z8-Ar, appearing yellowish and light gray colors, respectively (Fig. S1). These inexpensive surfactants are able to reduce the ZIF-8 particle size with homogeneous distribution in solution. As shown in the transmission electron microscopy (TEM) images, the Fe-Z8-Ar is composed of nanoparticles about 40 nm in diameter (Fig. 1b), similar to that of ZIF-8 prepared



**Fig. 2.** (a) XRD patterns of C-Z8, C-Fe-Z8-Ar and C-Fe-Z8-air. (b) High resolution Fe 2p XPS spectra of C-Fe-Z8-Ar and C-Fe-Z8-air. (c)–(e) TEM images of C-Z8 (c), C-Fe-Z8-Ar (d), and C-Fe-Z8-air (e). (f)–(h) Elemental mapping of C-Fe-Z8-Ar in the scanning TEM mode of the corresponding region in (f).

under the identical conditions but without Fe doping (Fig. S2). However, inhomogeneity features are found in Fe-Z8-air, suggested by a broader distribution of particle sizes in TEM images (Fig. 1c). X-ray diffraction (XRD) patterns indicate that these two Fe-doped ZIF-8 samples have similar crystal structure to the undoped ZIF-8 (Fig. 1d), suggesting Fe doping at such a low level (5 mol%) does not affect the ZIF-8 structure.

The Fe oxidation states in the two Fe-doped ZIF-8 samples are studied by X-ray photoelectron spectroscopy (XPS, Fig. 1e). Fe-Z8-Ar shows a main peak at 709.9 eV and a satellite peak  $\sim$ 715 eV, corresponding to exclusive divalent Fe. However, the main peak of Fe-Z8-air shifts to 710.9 eV accompanied by a more notable shift of the satellite peak to 718 eV. These features are indicative of trivalent Fe, which are very similar to that of Fe (III) oxide/hydroxide reported in previous studies [44,45]. In addition, the small hump observed at 705 eV for the Fe-Z8-air sample is related to Fe 2p 3/2 due to trace of the metallic aggregates. The comparison suggests that ambient air leads to a variety of iron species ranging from metallic Fe to Fe (III) during the preparation process for Fe-doped ZIF-8.

In principle,  $\text{Fe}^{2+}$  can form four fold coordinated tetrahedral complexes with imidazole-type ligands [46,47]. The similar local coordination environment and isovalent nature make  $\text{Fe}^{2+}$  substitution for  $\text{Zn}^{2+}$  in ZIF-8 favorable. Homogenous Fe distribution in Fe-Z8-Ar is further confirmed by elemental mapping (Fig. 1(f)–(h)). Such Fe substitution effect was also observed by Tian et al. using Mössbauer spectroscopy to study ZIF-8 that is in contact with Fe (II) complexes during an ion exchange process in aqueous solution [48].

### 3.2. Synthesis of Fe-N-C catalysts derived from Fe-doped ZIF-8

Subsequently, the Fe-N-C catalysts are prepared by a heat treatment of the ZIF nanoparticles at 1000 °C in Ar atmosphere. Catalysts derived from undoped ZIF-8, Fe-Z8-Ar and Fe-Z8-air are denoted as C-Z8, C-Fe-Z8-Ar, and C-Fe-Z8-air, respectively. Remarkable structural similarity is found for C-Z8 and C-Fe-Z8-Ar. In the XRD patterns (Fig. 2a), both samples only show two broad peaks centered at 25° and 43° corresponding to partially

graphitized carbon. XPS analysis (Fig. 2b) further indicates divalent Fe is exclusively found in C-Fe-Z8-Ar with a single peak at  $\sim$ 710 eV. The divalent nature suggests strong interaction between the highly dispersed Fe and the N-doped carbon [49,50]. This result agrees with the porous carbon morphology observed in the TEM images (Fig. 2c and d). Furthermore, elemental qualification determined by both Vario EL Elemental analyzer (Table S1) and XPS (Table S2) indicated that, different from C-Z8 (virtually metal-free N doped carbon), C-Fe-Z8-Ar does contain 3.35 wt% Fe (Table S1). In addition, despite of the absence of observable Fe agglomerates, Fe is further verified in the C-Fe-Z8-Ar sample by elemental mapping, showing uniform distribution of Fe on N-doped porous carbon (Fig. 2(f)–(h)). In MOF precursor, Fe-Z8-Ar, Fe is coordinated by four N atoms from 2-methylimidazole (Fig. 1a), which is favorable for the formation of the uniform  $\text{FeN}_x$  sites into the porous carbon matrix after a simple thermal conversion [10,35]. It should be noted that the residual Zn is negligible below 0.05 at% for Fe-based catalysts. The XPS survey spectra of C-Z8, C-Fe-Z8-Ar, and C-Fe-Z8-air also were compared in Fig. S3. It should be not an issue to impair the long term stability in fuel cells due to the contamination of ionomer and membrane with  $\text{Zn}^{2+}$ . Therefore, the additional acid leaching treatment is not needed for such-synthesized catalysts.

Similar to C-Fe-Z8-Ar, the C-Fe-Z8-air contains 4 wt% Fe, which was prepared from the Fe-doped ZIF-8 without Ar protection. Considerable amount of metallic Fe, iron carbide and trace Fe(II) oxide were clearly detected by the associated peaks in XRD patterns. The XPS peak at 707 eV in Fig. 2(b) is also indicative of metallic Fe in the C-Fe-Z8-air sample. These are in agreement with the crystalline Fe containing particles detected by TEM (Fig. 2e). A morphology comparison determined by SEM images between C-Fe-Z8-Ar and the C-Fe-Z8-air presents in Fig. 3.

Unlike amorphous or partially graphitized structure observed with the C-Fe-Z8-Ar, there appear dominant carbon nanotube-like structures in C-Fe-Z8-air. The formation of carbon nanotubes is likely catalyzed by the generated metallic Fe or  $\text{FeC}_x$  in catalysts. Usually, the formation of highly graphitized nanocarbon such as carbon tube, onion-like carbon, and graphene-sheet structures were extensively observed in high-performance Fe-N-C catalysts

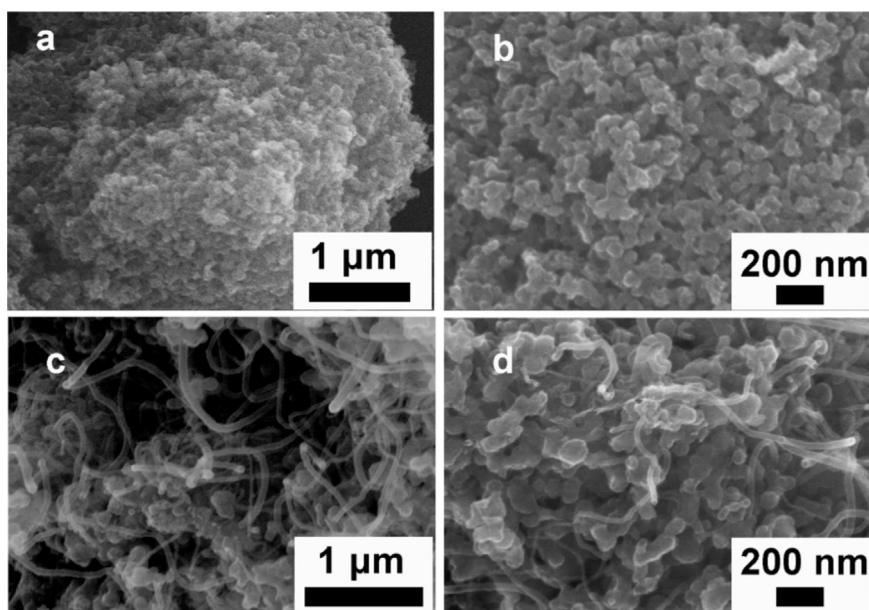


Fig. 3. SEM images of C-Fe-Z8-Ar (a,b) and C-Fe-Z8-air (c,d).

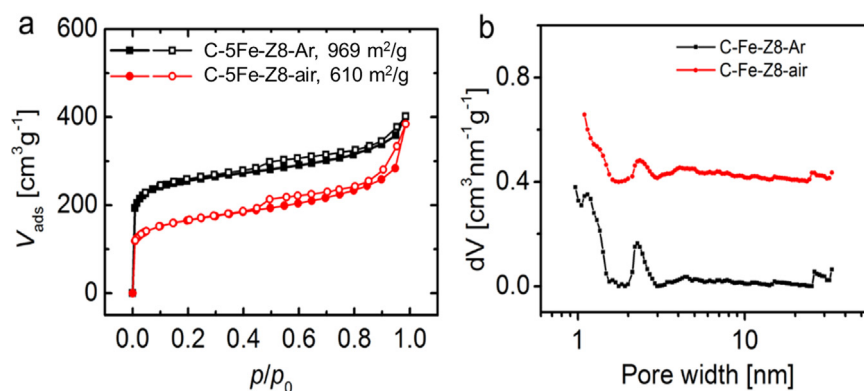


Fig. 4. Nitrogen adsorption-desorption isotherms (a) and pore size distribution (b) for C-Fe-Z8-Ar and C-Fe-Z8-air catalysts.

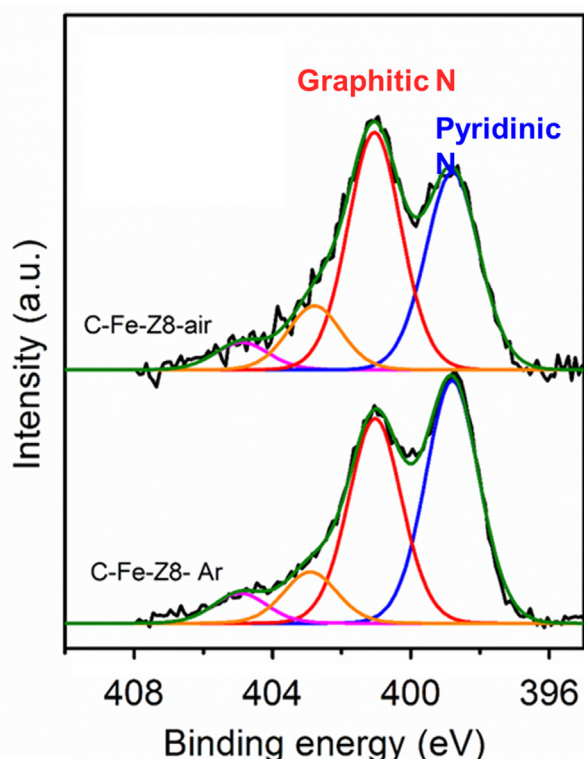


Fig. 5. High resolution N 1s XPS spectra of C-Fe-Z8-Ar and C-Fe-Z8-air.

[24,51–53]. Initially, we thought that these highly graphitized nanocarbons in catalysts dictate active site generation and was directly linked to the enhanced ORR activity and stability. Oppositely, in this work, dominant amorphous or partially graphitized carbon with uniform distributions of N and Fe were found in the C-Fe-Z8-Ar catalyst showing even higher ORR activity ( $E_{1/2}$  up to 0.82 V vs. RHE, to be discussed later) and stability relative to graphitized nanocarbon-rich Fe-N-C catalysts (0.80 V). Thus, it seems that the highly graphitized carbon structures in the Fe-N-C catalysts are not necessary for generating high ORR activity. It should be noted that dominant agglomerated iron compounds accompanying with highly graphitized nanocarbons in catalysts suggest a non-uniform dispersion of active iron sites, which is controversial to the intention to increase density of active sites. Interestingly, formation of carbon nanotubes was also observed in the pyrolysis product of Fe-doped ZIF-8 by Su et al. showing

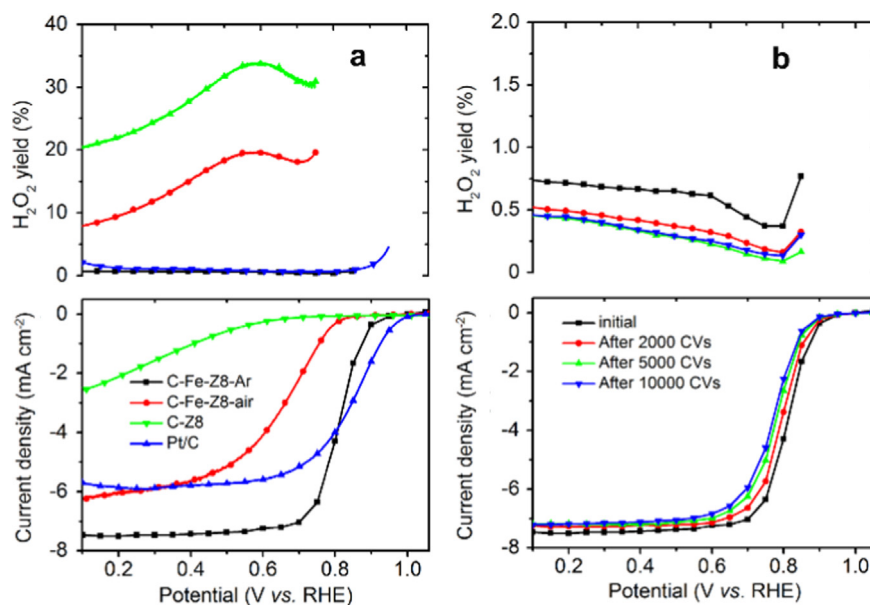
insufficient ORR activity [43]. The Fe-doped ZIF-8 in their work was obtained from a solvothermal reaction without deliberately eliminating the ambient air, very similar to the Fe-Z8-air in this work. As shown in Fig. 4a, the Brunauer–Emmett–Teller (BET) surface area of C-Fe-Z8 air is also much lower (610 m<sup>2</sup>/g) compared to those of C-Fe-Z8-Ar (969 m<sup>2</sup>/g) and Fe-free C-Z8 (1049 m<sup>2</sup>/g). As for Fe-N-C catalysts, larger surface areas, higher ORR activity [54]. Therefore, controlling catalyst morphology with improved surface areas is also critical for enhancing catalytic performance. In addition, the C-Fe-Z8-Ar processes dominant microporous structure compared to C-Fe-Z8-air (Fig. 4b), which is unfavorable for mass transport during the ORR [23,55].

The nature of the nitrogen dopant was believed to be crucial for the ORR activity of Fe-N-C type catalysts. The N 1s XPS spectra (Fig. 5) suggest that both C-Fe-Z8-Ar and C-Fe-Z8-air contain dominant graphitic (401.8 eV) and pyridinic (398.7 eV) N along with small fraction of oxidized nitrogen (403–405 eV). The pyridinic N also includes contribution from the N bonded to metal which should appear at ~399.2 eV and largely overlaps with the pyridinic N in the measured spectra. Importantly, the C-Fe-Z8-Ar sample contains higher fraction of pyridinic N compared to the C-Fe-Z8-air. In addition, the total nitrogen content of C-Fe-Z8-Ar (6.9 wt%) is also more than twice of C-Fe-Z8-air (2.8 wt%, Table S1). Considering the much higher Fe dispersion with exclusively divalent state, the higher amount of pyridinic N in the C-Fe-Z8-Ar is also a strong indication of more Fe–N bonding in the catalyst.

### 3.3. Electrochemical activity and stability

Fig. 6a shows the ORR polarization plots and H<sub>2</sub>O<sub>2</sub> yield for the four catalysts measured on a RRDE. Onset potential is defined as the potential that is able to generate a current density of 0.1 mA/cm<sup>2</sup> during the ORR based on the recorded steady state polarization plots. For the best performing C-Fe-Z8-Ar, the onset potential ( $E_{on}$ ) and half-wave potential ( $E_{1/2}$ ) are determined to be 0.95 and 0.82 V, respectively. Compared to the commercial Pt/XC-72 (20 wt% Pt) catalyst, the newly developed C-Fe-Z8-Ar only shows slightly negative  $E_{1/2}$  value by ~40 mV in more challenging acidic electrolyte and significantly reduce the current gap between NPMC and Pt/C [56]. In addition, these values are much higher than those of C-Fe-Z8-air (0.84 and 0.65 V for  $E_{on}$  and  $E_{1/2}$ ) without well-dispersed atomic iron distribution. Furthermore, the calculated mass activity of C-Fe-Z8-Ar is higher than that of C-Fe-Z8-air (Fig. S4). Tafel slopes for various samples were determined in relatively high potential ranges (Fig. S4). The value for Pt/C is –60 mV/decade. Both C-Fe-Z8-Ar and C-Fe-Z8-air have similar slopes of –70 and –77 mV/decade. As for the iron-free C-Z8, the





**Fig. 6.** (a) Steady-state polarization curves (bottom) and H<sub>2</sub>O<sub>2</sub> yield (top) of the ORR catalysts derived from ZIF-8, Fe-doped ZIF-8 and E-TEK Pt/XC-72 with 20 wt% Pt loading. (b) Steady-state polarization curves (bottom) and H<sub>2</sub>O<sub>2</sub> yield (top) of C-Fe-Z8-Ar after different cyclic voltammetry cycles. All tests are carried out on a rotating ring disk electrode (RRDE) rotating at 1600 rpm in O<sub>2</sub>-saturated 0.1 M HClO<sub>4</sub>. The non-precious metal catalysts loading is controlled at 0.56 mg/cm<sup>2</sup>. The Pt/C loading is controlled at 20 μgPt/cm<sup>2</sup>.

value of slope is determined to be  $-236$  mV/decade, indicating, instead of first electron transfer ( $-118$  mV/decade) and possible immigration of intermediates ( $-59$  mV/decade), O<sub>2</sub> adsorption become a new rate determined step [57]. Thus, C-Fe-Z8-Ar is much more active than C-Fe-Z8-air. As expected, both Fe containing samples exhibit higher ORR activity than the Fe-free C-Z8 sample, which agrees with the general accepted conclusion that Fe is indispensable for good ORR activity in acidic media [32,58]. Noteworthy, due to the very porous structures for the catalyst, the measured limiting current densities are often higher than those calculated theoretical value by using flat disk 2D electrode. This is very common for good catalysts to generate increased limiting current due to high-surface areas and increased active sites in 3D electrodes.

Regarding the insignificant loading dependence of ORR activity as shown in Fig. S5a, it is due to that the C-Fe-Z8-Ar catalyst is highly active and contain high-density active sites in the 3D porous electrode, capable of providing sufficient active sites even at a low loading. Four-electron selectivity is another criterion for ORR catalysts. An ideal ORR catalyst for PEFCs should catalyze O<sub>2</sub> to H<sub>2</sub>O via a  $4e^-$  reduction pathway without H<sub>2</sub>O<sub>2</sub> formation. Calculated from RRDE data using Eq. (1), the electron transfer number is 3.98 for the C-Fe-Z8-Ar in the whole potential region from 0.2 to 0.8 V, indicating extremely low H<sub>2</sub>O<sub>2</sub> yield below 1%. Importantly, there is no monotonic variation trend of H<sub>2</sub>O<sub>2</sub> yield with catalyst loadings (Fig. S5b), indicating a real direct 4-electron reduction, rather than a stepwise  $2e^- + 2e^-$  reduction mechanism [59–61]. Thus, a more noticeable advantage of the catalyst is the extremely low H<sub>2</sub>O<sub>2</sub> yield ( $< 1\%$ ) without loading dependence. H<sub>2</sub>O<sub>2</sub> is especially undesirable for Fe-based ORR catalyst due to the possible Fenton reaction in the presence of Fe<sup>2+</sup>, which will damage the Nafion® membrane in PEFCs [62]. For the C-Fe-Z8-air, the electron transfer number is between 3.35 and 3.81 in the same potential region. The corresponding H<sub>2</sub>O<sub>2</sub> yield is as high as 20%.

In this work, the optimal Fe content is determined around 5 at% in Fe-doped ZIF-8. We also prepared catalysts with nominal Fe content up to 10 at% and down to 1 at%. But the yielded ORR activities are inferior to the catalyst with optimal 5 wt% Fe (Fig. S6). Meanwhile, we found significant Fe-containing aggregates in the

TEM image of C-10Fe-Z8-Ar (Fig. S7a), which was further verified as Fe and Fe<sub>3</sub>C by analyzing its XRD pattern (Fig. S7b). This is in agreement with the main conclusion of the paper, which high atomic Fe dispersion is crucial to achieve good ORR performance. Furthermore, the best performing C-Fe-Z8-Ar catalyst also shows very high methanol tolerance and sufficient stability. As shown in Fig. S8, introducing methanol has little impact on the ORR peak of the C-Fe-Z8-Ar catalyst. Importantly, the long term stability is assessed by the ORR polarization curves before and after different potential cycles in the range of 0.6–1.0 V. Fig. 6b indicated that its  $E_{1/2}$  is only negatively shifted by 40 mV after 10,000 cycles. The slight degradation may be attributed to loss of active iron sites hosted in the carbon matrix caused by carbon oxidation [63]. Interestingly, the H<sub>2</sub>O<sub>2</sub> yield is further reduced to less than 0.5% during the potential cycling test. Therefore, compared to the state-of-the-art Fe-N-C catalysts, the newly developed C-Fe-Z8-Ar from MOFs demonstrates a very high half-wave potential and significantly reduced H<sub>2</sub>O<sub>2</sub> yield lower than 1% [3,12,50,64]. A comprehensive comparison data with other Fe-N-C catalysts are summarized in Table S3.

### 3.4. Role of Fe(II) in enhancing activity

The enhanced ORR activity of the C-Fe-Z8-Ar catalyst is mainly due to the increased density of active sites associated with high Fe dispersion at atomic level. Particularly, the C-Fe-Z8-Ar catalyst contains homogeneously distributed Fe without observable Fe-containing particles, capable of maximizing the ORR active Fe sites. The highly porous morphology also allows facile access to the Fe sites by O<sub>2</sub>. The structure of Fe-doped ZIF-8 illustrated in Fig. 1 (a) is ideal to achieve uniform Fe distribution in the catalyst during the thermal conversion. In this structure, Fe is homogeneously dispersed by Zn, in which the Zn species itself is removable during the thermal conversion. Thus, aggregation of Fe, which frequently occurs at high temperature treatment, is effectively minimized through using Fe-doped ZIF-8 MOF precursors.

It should be noted that, to achieve homogeneous Fe doping in ZIF-8, Fe<sup>2+</sup> is preferred compared to Fe<sup>3+</sup>, because Fe<sup>2+</sup> and Zn<sup>2+</sup> are isovalent with similar ion radius (cation radii: Fe<sup>2+</sup> 62 pm and

Zn<sup>2+</sup> 60 pm) [65] and stabilization constants with 2-methylimidazole [48]. As a result, tetrahedral FeN<sub>4</sub> structure has been extensively found in complexes composed of Fe<sup>2+</sup> and imidazole type ligands [46,47,66]. However, Fe doping in ZIF-8 is carried out in a weakly alkaline environment (pH ~ 10), in which Fe(II) can be easily oxidized to Fe(III) by ambient O<sub>2</sub>. Fe<sup>3+</sup> has much smaller radius (49–55 pm) and typically adopts six fold coordination structure in imidazole complexes [66]. Thus, Fe<sup>3+</sup> substitution for Zn<sup>2+</sup> in ZIF-8 will cause significant structural distortion, which is less stable. Furthermore, Fe<sup>3+</sup> shows much higher hydrolysis tendency compared to the divalent cations ( $K_{sp}$ : 10<sup>−38.5</sup> for Fe(OH)<sub>3</sub>, 10<sup>−16.5</sup> for Zn(OH)<sub>2</sub> and 10<sup>−16.3</sup> for Fe(OH)<sub>2</sub>) [65]. Thus, in a weakly alkaline solution, hydrolysis of Fe<sup>3+</sup> is highly favorable, as clearly evidenced by the detection of Fe(III) oxide/hydroxide in XPS (Fig. 1(e)). The inhomogeneity observed in the TEM image (Fig. 1(c)) is also very likely due to the hydrolysis of Fe<sup>3+</sup>.

Therefore, an O<sub>2</sub>-free environment is required for stabilizing Fe<sup>2+</sup> to facilitate homogeneous Fe substitution for Zn in ZIF-8, which is crucial for uniform Fe distribution in the catalyst. Otherwise, oxidation of Fe<sup>2+</sup> to Fe<sup>3+</sup> and the following hydrolysis of Fe<sup>3+</sup> will result in Fe aggregation in the derived Fe-N-C catalyst. This key factor has been overlooked in previous studies using ZIF-8 precursors, which yielded a Fe-N-C catalyst with moderate ORR performance similar to C-Fe-Z8-air in this work [33]. The negative impact of O<sub>2</sub>, however, is less critical in Fe/ZIF-8 composites prepared by ball milling when stable N-chelated Fe complexes are used. In that case, the driving force to disperse Fe in ZIF-8 is physical mixing rather than chemical Fe substitution for Zn [16,35,36].

#### 4. Conclusions

In summary, starting with homogeneously Fe-doped ZIF-8, a highly active and stable Fe-N-C catalyst with highly dispersed Fe sites is obtained via facile one-step thermal conversion without requirements of post-leaching and additional heating treatments. The simple synthesis procedure is low-cost and allow for catalyst production in large scale. A significant innovation during the synthesis is that, for the first time, Fe was successfully doped into ZIF-8 by partially replacing Zn ions, when maintaining divalent during the synthesis process. During the thermal treatments, nitrogen and iron coordination (FeN<sub>4</sub>) containing ZIF-8 framework directly converts into N-doped carbon with uniform Fe distribution without any agglomerated iron particles. The zinc species is eliminated due to its high volatility, yielding a highly porous carbon morphology. Notably, the control of an O<sub>2</sub>-free environment for stabilizing Fe(II), rather than forming Fe(III), during the preparation of Fe-doped ZIF-8 precursor was found to be extremely crucial for generating uniform Fe distribution at atomic level. This strategy effectively prevents formation of inactive iron aggregates.

Importantly, the new Fe-N-C catalyst exhibits very high ORR activity in terms of its half-wave potential (as high as 0.82 V vs. RHE) and extremely low H<sub>2</sub>O<sub>2</sub> yield (< 1%) as well as sufficient potential cycling stability in the more challenging acidic electrolyte. This effort significantly reduces the activity gap between current NMPs and Pt/C catalysts in more challenging acidic media. The membrane electrode assembly (MEA) fabrication using the newly developed Fe-N-C catalyst is ongoing for PEFC applications, which will be reported in near future. Noteworthy, an important finding from this work is that highly graphitized nanocarbons once extensively observed in NMPs are not necessities for high ORR activity, which differs to our previous observation toward the ORR electrocatalysis.

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#### Appendix A. Supporting information

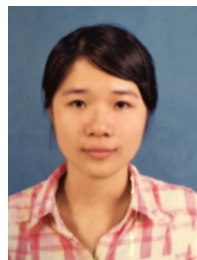
Supplementary data associated with this article can be found in the online version at <http://dx.doi.org/10.1016/j.nanoen.2016.04.042>.

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